Utkarsh Sarvjeet Tiwari

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Education

- **Shah and Anchor Kutchhi Engineering College**
  B. TECH in Electronics and Computer Science Engineering (Third year, Semester 6)
  Graduation: 2027 **Sept 2023 – Present**
  Mumbai, Maharashtra

Certifications

- **Linux Essentials – Cisco Networking Academy**
- **Embedded System Developer – AICTE Virtual Internship (Eduskills Foundation)**

Summary

Electronics & Computer Science undergraduate specializing in AI/ML, NLP, and Quantum Computing with hands-on experience in product development and research internships. Finalist in multiple national hackathons seeking AI/Software Engineering roles.

Technical Skills

- **Programming Languages:** Python, C, C++, JavaScript.
- **AI & Machine Learning:** Machine Learning, NLP, Sentence Transformers, TensorFlow, Keras, Scikit-learn.
- **Data & Visualization:** NumPy, Pandas, Matplotlib, Data Preprocessing, Model Evaluation.
- **Web & Frameworks:** Flask, Node.js, React.js (JSX)
- **Databases & Tools:** MySQL, MongoDB, Git, VS Code, XAMPP, OpenCV
- **Quantum (Basics):** Qiskit, PennyLane

Experience

- **Research Intern (AI/ML) - NLIP Lab, IIT Hyderabad** Feb 2026 - Present
  - Contributing to development of an **AI-based product/system**.
  - Building **data pipelines and preprocessing workflows** for real-world datasets
  - Performing **model evaluation and optimization** to improve accuracy and response time
  - Collaborating with research scholars and faculty on **system architecture and experimentation**

- **AI/ML Intern - Aryahs World Infotech** Dec 2025 - Jan 2026
  - Developed **WhatsApp automation system** for customer interaction and workflow management
  - Built **ML-based symptom classifier** mapping user inputs to probable medical conditions
  - Implemented **automated email generation system** for context-aware communication
  - Contributed to **full-stack product features** using Python, Flask, MySQL & JavaScript
  - Worked with real-time datasets, testing, and Agile sprint collaboration

Projects

- **Quantum Computing Research Project (Industry-Defined Problem by Cybranex Pvt. Ltd.)**
  (*Under Faculty Mentorship – **Ongoing)***
  - Designed and trained quantum autoencoder models for data compression and feature extraction as part of an industry-defined research problem under faculty mentorship.
  - Simulating **quantum circuits** and exploring hybrid quantum–classical architectures for dimensionality reduction.
  - Conducting research and documentation under faculty mentorship as part of a **college–industry collaboration program**.
  - **Tech:** Qiskit, Python, NumPy, Matplotlib

- **Local Voice AI Agent** (GitHub)**:** **August-2025**
  - Developed a Windows-first voice chat app using local AI models (Gemma 3 via Ollama) with real-time STT/TTS pipeline.
  - Implemented conversational memory, retry logic, and performance metrics while defaulting to 1B model for minimal downloads.
  - Created configurable CLI with tuning parameters and multiple deployment modes (local/LAN/public sharing).
  - Technologies: Python, Ollama, Gradio, WebRTC, YAML

- **Ecommerce Web Application – Artha Mart** (Github): **June-2025**
  - Developed a group-based E-commerce web application providing a seamless online shopping experience.
  - Designed and implemented the **MySQL** database schema for users, products, orders, and cart management.
  - Built backend logic in **Flask** to handle user authentication, product management, cart operations, and order processing.
  - Wrote secure and efficient SQL queries for data retrieval and manipulation.
  - Ensured smooth communication between the Flask application and the MySQL database.

- **Harry Potter Invisibility Cloak using OpenCV and Python** (Github): **March-2025**
  - Developed a computer vision project simulating the invisibility effect using color detection and HSV masking in real-time video.
  - Utilized **OpenCV** and **NumPy** libraries to capture background, apply color masks, and blend frames to produce the invisibility illusion.
  - Included real-time trackbar tuning and a demo video demonstration.
  - Hosted code and video on GitHub.

- **Smart Home Automation System** (Github): **Dec – 2024**
  - Device control & energy monitoring simulator.

Hackathon & Innovation Experience

- IIT ISM Dhanbad Finalist
- HackWithUP Shortlisted
- HackSetu Grand Finale
- E^2 Calonics Paper Selection

Achievements

- Led 12 campus ambassadors at ISRC
- Finalist in multiple national AI hackathons